

TONIGHT

The price of being wrong

A score, a threshold and the price of each mistake. Then the quiz, two ways models lie, and an hour on how I work with Claude.

Data Analysis & Machine Learning for Business

Andrey Shumilin

So which one is better?

LOGISTIC REGRESSION

As it comes

12

of 66 caught

96.9%

accuracy

LOGISTIC REGRESSION

With class weights

57

of 66 caught

272

false alarms

Accuracy could not say. **Tonight we answer with money, through your HW2 cost ratio.**

From a score to a decision you can defend.

01 Matrix, threshold, ROC and money

18:12, Demo 1

02 The quiz

19:00, in person

03 Why models lie: leakage and the lucky split

19:40, Demo 2

04 How I work with Claude, live

20:00, one hour

Demo 01

Session 5's boosting, class weights off.
Matrix, threshold, ROC and money.

THE COST OF THE EVENING

cost = 10 × missed + false alarms

At 19:00 we stop wherever we are.

- 1 Model and score: **we move the threshold by hand**
- 2 Confusion matrix at 0.5: 52 missed, 14 caught
- 3 Which mistake hurts more: flag everyone, flag one firm
- 4 Slide the dial: $0.5 \rightarrow 0.3 \rightarrow 0.1 \rightarrow 0.04$
- 5 ROC and AUC: toy 0.83, boosting 0.949
- 6 Expected cost: best threshold **0.04**, 295 vs 528
- 7 Ratios 0.5 / 5 / 10 / 25, and class weights back on
- 8 **19:00, the quiz.** The rest goes home

Ask Your HW2 ratio is in the README. From the curves alone, where does your threshold go?

Four cells, two honest rates.

	WE FLAGGED	WE CLEARED
TRULY BANKRUPT	TP caught	FN missed, the expensive one
TRULY HEALTHY	FP false alarm	TN correctly cleared

RECALL, THE TOP ROW

$$TP / (TP + FN)$$

Of all bankrupt firms, how many did we catch?

PRECISION, THE LEFT COLUMN

$$TP / (TP + FP)$$

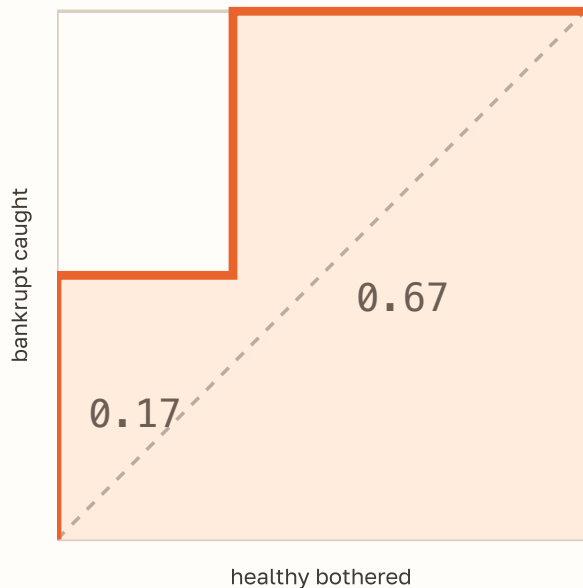
Of all our flags, how many were real?

scikit-learn prints it upside down: healthy first, top-left is TN.

Accuracy mixes all four cells into one number. **At 3% bankrupt firms, it mixes them into a lie.**

AUC grades the order. The threshold makes the cut.

TOY: 2 BANKRUPT, 3 HEALTHY



TWO READINGS OF 0.83

The area: $0.17 + 0.67$. Or the pairs: **5 of 6** bankrupt-healthy pairs ranked the right way.

0.949

boosting

0.45

random scores

1.0

perfect corner

THE LEFT EDGE

5% of healthy firms bothered is already **99 false alarms**. There the model catches 70%.

The order is good. **The cut at 0.5 is in the wrong place.**

The default 0.5 costs 1.8 times as much.

BEST THRESHOLD

0.04

cost **295**

DEFAULT

0.5

cost **528**

FOUR NUMBERS FOR HW2 PART 5, AT THE OPTIMUM

9%

firms flagged

74%

recall

28%

precision

295

vs 528 at 0.5

YOUR RATIO, YOUR THRESHOLD

COST RATIO

BEST THRESHOLD

0.5

0.82

5

0.15

10

0.04

25

0.01

A dearer miss means a lower threshold. Below 1, a false alarm is dearer: the hotels in HW2.

72% of our flags are false alarms, **and that is the cheapest answer.**

Phones out, books closed.

Part A 10 questions on the basics. **Enough to pass.**

Part B 4 harder questions for a higher grade, two with short written answers.

QUIZ FORM



The form closes at **19:30**

Two liars in every test number.

01

Leakage

A column or a step secretly knows the answer. The model shines until it meets real life.

Too good to be true is true.

02

The lucky split

One test number is itself random. With 3% positives the test part holds a few dozen bankrupt firms.

Reshuffle the split, and the AUC moves.

Both are caught by **process**, not by staring at the model.

Demo 02

A crude leak, a subtle one, and the lucky split.

ONE QUESTION PER COLUMN

Known before the outcome, or only after?

This is HW2 Part 0. If you did it already, reopen your list tonight.

- | | |
|---|--|
| 1 | Crude leak: audit_flag built from the target. Test AUC 1.0000 |
| 2 | Boutique leak: 240 firms, 500 indicators of pure noise. Select then split, or split then select |
| 3 | Five folds, AUC 0.84 to 0.96. CV to choose, the test once |

Ask 500 noise indicators, 240 firms. What AUC will the tempting pipeline get? Bets.

Selection is part of the model.

THE TEMPTING ORDER

Select on all firms, then
split

MEAN AUC, 10 SPLITS

0.86

single splits up to 0.96

THE HONEST ORDER

Split, then select on train
only

MEAN AUC, 10 SPLITS

0.54

as noise deserves

THE LUCKY SPLIT

Honest splits, from noise

0.38 to 0.80

4 bankrupt firms in each test part. **CV**
to choose, test once.

Everything that computes something from the data **lives inside the training part.**

Before you believe a model.

01 Split first. **Everything that learns** stays inside the training part.

02 The gap between train and test is checked.

03 The metric fits the balance: AUC, recall and precision, not accuracy.

04 Threshold chosen by **costs**, its price shown against 0.5.

05 Every column known before the outcome. Too good to be true means a leak hunt.

06 Small data: CV to choose, the test touched once.

HOME READING, 03-METRICS-MAP.IPYNB

It starts from the business goal: a model metric is its stand-in, and **when none fits, you write your own**. Two tables at the bottom list every metric we used, with when it misleads.

Six things I hand to AI.

01

This course

notebooks, slides, numbers checked against the data

02

Skills

reusable instructions for a recurring job

03

Deep research

a question in, a sourced report out

04

Computer and browser use

the assistant clicks through sites and apps

05

Apps in Xcode

built-in support for app development

06

Live: a financial report

Moscow Exchange export → report.
Design + coding, after a short intro to the tools

Think, formulate, **check**.

Seven ideas to keep.

01 A score is a probability only if it is calibrated. A threshold turns it into a decision.

03 The ROC curve and AUC grade the order. The threshold makes the cut.

05 Too good to be true is true. Ask every column when it becomes known.

07 Start from the business goal when you choose a metric.

02 Four cells, two honest rates, a price on each mistake.

04 The threshold is a money dial: 1.8 times the cost was on the table.

06 CV to choose, the test once.

NEXT WEDNESDAY, THE FINALE

We cluster thousands of real bank complaints by meaning. HW2 is due **October 21, 18:00**: questions go to the chat before the deadline.